

Lab 2 – SciLab

Introduction

This lab will familiarize the student with the capabilities and basic use of SciLab which is a freely available calculation program. It is similar to the much more expensive commercial tool MATLAB. MATLAB is widely used in industry and has advanced capabilities in graphics plotting and code generation that go well beyond the capabilities of SciLab.

Part I: Basic SciLab

There are downloads and many tutorials available on the SciLab website: www.scilab.org

The supported platforms are Windows (multiple versions), Linux and Mac OS.

1. Load the version appropriate to your computer
2. Download the “SciLab - Tutorial.pdf” document in Blackboard
3. Perform the first 23 pages of the tutorial
 - Stop at the paragraph titled “Additional Information for Matrices and Vectors”
 - We will cover this section (and the rest) during Part II
4. Develop the program specified in this practical including a short test for it
5. Create a single PDF document with your program and the console output from a test run

Program Psuedo Code

This psuedo code just describes the math transforms necessary to implement

- polar to Cartesian transforms and
- Cartesian to polar transforms

Cartesian coordinates are represented by a complex number. Code needs to be written for each item and tested **all in one file**.

1. Define any constants you need
 - Make sure to include: $\%j = \text{sqrt}(-1)$ so that the symbol **j** used in EE notation can substitute for **i** that is used in conventional mathematical notation
2. Write a SciLab function **polar2cart(m,a)** (or any name you choose)
 - The function should take 2 arguments and implement the polar to Cartesian conversion resulting in a complex number, **x**, returned from the function
 - The **m** is for magnitude and **a** is for the angle of the polar form of the function arguments, or

$$x = m \cos(a) + \hat{j} m \sin(a)$$
3. Write a SciLab function **cart2polar(x)** (or, again, your choice for function name)
 1. This function should take a complex number, **x**, and convert it to polar form
 2. The resultant value of the function should be in a 2 element row matrix of the form **[m a]** where

$$m = \sqrt{\Re^2(x) + \Im^2(x)} \quad \text{and} \quad a = \tan^{-1} \left[\frac{\Im(x)}{\Re(x)} \right]$$

where $\Re(x)$ is the “real” part and $\Im(x)$ is the “imaginary” part of **x**

4. Write a “main” program following these two functions that calls each of these functions with test argument(s) and displays the arguments and the result. Also display your name so the instructor knows who created the program.

Part II: Basic SciLab

A system of linear equations is defined in matrix form as:

$$Y = A * X, \text{ where}$$
$$A = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix} \text{ and } X = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

Continue the “SciLab - Tutorial.pdf” from Part I, starting off where you last left off (p23 with Matrices and Vectors). For this portion of the lab, you will write a SciLab program to perform an example matrix vector multiplication and solve the set of linear equations given.

1. Display your name
2. Define A and X as given above
3. Display A and X with labels
4. Solve and display Y via matrix-vector multiplication
5. Invert matrix A and display (this is A^{-1})
6. Solve and display $A^{-1} * Y$ (this should be the same as X)
7. Create a single PDF document with your program and the console output

Lab Submission

Turn in both parts as a single PDF submission to Lab2.